



Classification - Part 3

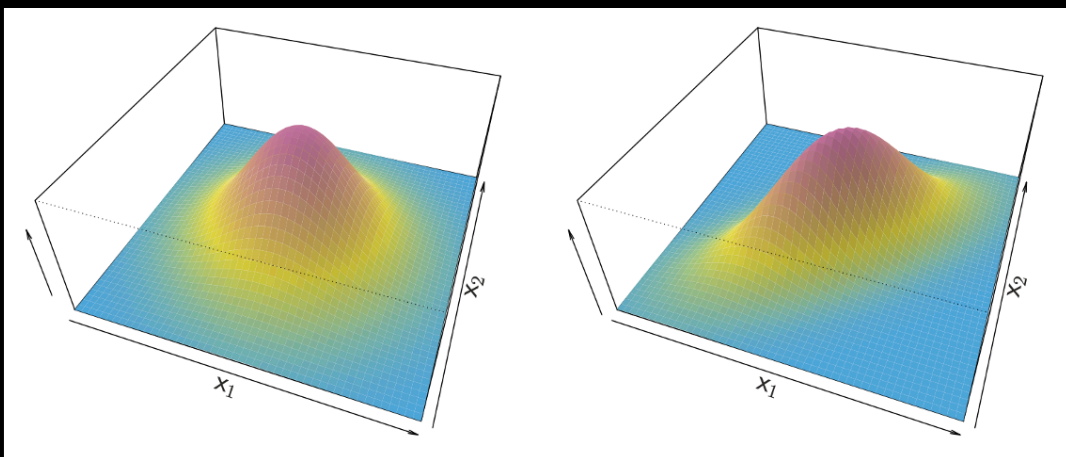
QDA, Naive Bayes, Poisson Regression



LDA for $p > 1$

Assume $X = X_1, \dots, X_p$ is drawn from a multivariate Gaussian with class-specific mean vector and common covariance matrix Σ :

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$





LDA for $p > 1$: Using the Bayes Classifier

The Bayes classifier assigns an observation $x = (x_1, \dots, x_p)$ to the class k for which the following discriminant is the largest:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \mu_k$$

We need to estimate the unknown parameters $\mu_1, \dots, \mu_k, \Sigma, \pi_1, \dots, \pi_k$

We can generalize the same formulas for $p = 1$



What is Σ ?

Multivariate normal

$$p(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right)$$

Multivariate normal (p=2)

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2) \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

$$\rho(X, Y) = \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \Rightarrow \text{Cov}(X, Y) = \rho \sigma_X \sigma_Y$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{bmatrix}$$

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp \left(-\frac{z}{2(1 - \rho^2)} \right), \quad x, y \in \mathbb{R}$$

$$z = \frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}$$



Common Variance Assumption in LDA

In LDA, σ and Σ are not a function of the class k

The variance is *common* across all classes

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

$$\hat{\delta}_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \mu_k$$



Next Time

QDA

Naive Bayes

(Poisson Regression)



Quadratic Discriminant Analysis (QDA)

Like LDA: assume that the observations are drawn from a multi-variate Gaussian distribution, and use Bayes classifier

Unlike LDA: assume that the covariance matrix is not the same (common) across the K classes

That is, we need to estimate a different Σ_k for each class



Discriminant and Estimates in QDA

An observation x is assigned to the class that maximise the following (quadratic) discriminant:

$$\sigma_k(x) = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) - \frac{1}{2}\log |\Sigma_k| + \log \pi_k$$

Estimate of Σ_c for class c :

$$\frac{1}{n_c - 1} \sum_{y_i=c} (x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T$$

where n_c is the number of samples in class c



Naive Bayes

The problem in generative models is estimating $f_k(x)$, which is normally a p -dimensional distribution.

$$Pr(Y = k | X = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{l=1}^K \pi_l f_l(\mathbf{x})}$$

In LDA and QDA we assume this distribution is normal.

What if we assume that, within the class k , the p predictors are independent?

$$f_k(\mathbf{x}) = f_{k1}(x_1) \times f_{k2}(x_2) \times \cdots \times f_{kp}(x_p)$$



Naive Bayes

$$Pr(Y = k|X = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{l=1}^K \pi_l f_l(\mathbf{x})}$$

Assuming: $f_k(\mathbf{x}) = f_{k1}(x_1) \times f_{k2}(x_2) \times \cdots \times f_{kp}(x_p)$

Becomes:

$$Pr(Y = k|X = \mathbf{x}) = \frac{\pi_k \times f_{k1}(x_1) \times \cdots \times f_{kp}(x_p)}{\sum_{l=1}^P \pi_l \times f_{l1}(x_1) \times \cdots \times f_{lp}(x_p)}$$

Now, we need to estimate only the one-dimensional density functions $f_{ki}(x_i)$ using the available observations.



Estimating the Density Functions $f_{ki}(x_i)$ for Quantitative Predictors

Option 1

We can assume that within each class, the j -th predictor is drawn from a univariate normal distribution:

$$X_j|Y = k \sim N(\mu_{jk}, \sigma_{jk}^2)$$

Then use the same estimates of μ_{jk} and σ_{jk}^2 of QDA



Estimating the Density Functions $f_{ki}(x_i)$ for Quantitative Predictors

Option 2

Use a *non-parametric* estimate for $f_{kj}(x_j)$.

For example, make a histogram of the observations of the j -th predictor within each class k . Then, estimate $f_{kj}(x_j)$ as the fraction of the training observations in the k -th class that belong to the same histogram bin as x_j



Estimating the Density Functions $f_{ki}(x_i)$ for Qualitative Predictors

Simply count the proportion of observations for the j -th predictor corresponding to each class.

Example (100 observations):

Suppose the predictor j takes on values of 1, 2, and 3, in 32, 55, and 13 observations, respectively. Then:

$$f_{kj}(x_j) = \begin{cases} 0.32, & \text{if } x_j = 1 \\ 0.55, & \text{if } x_j = 2 \\ 0.13, & \text{if } x_j = 3 \end{cases}$$



Naive Bayes: Example

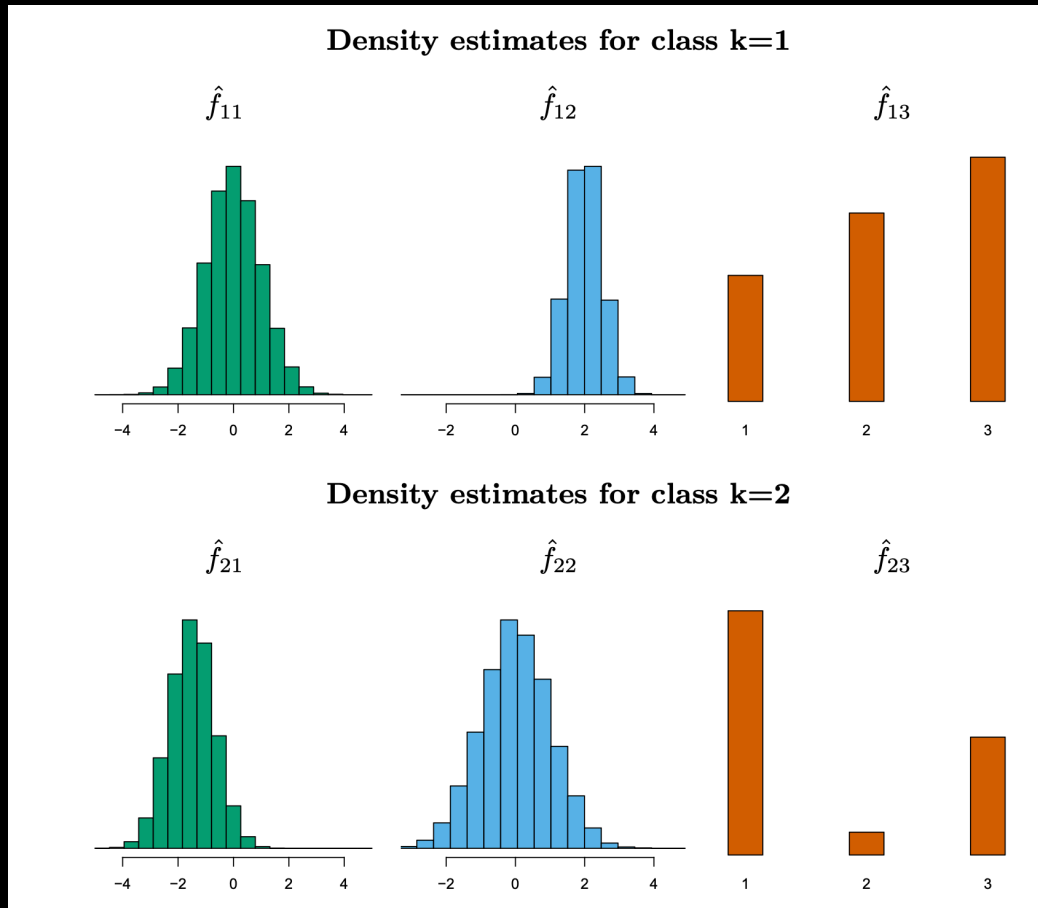
$p = 3$ predictors and $K = 2$ classes.

The first two predictors are quantitative, the third one qualitative with 3 levels.

We assume also that $\hat{\pi}_1 = \hat{\pi}_2 = 0.5$ (balanced classes)



Naive Bayes Example: Density Estimates





Evaluating the Performance of a (Binary) Classifier

Confusion Matrix

Accuracy

Precision

Recall

F1 Score

AUC



Confusion Matrix (Binary Classification)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- TP: correctly predicted positives
- TN: correctly predicted negatives
- FP: negatives incorrectly predicted as positive
- FN: positives incorrectly predicted as negative



Accuracy

Definition: Fraction of all predictions that are correct.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Interpretation:

“Out of all instances, how often is the model right?”

Simple and intuitive, but can be misleading with imbalanced classes.



Precision

Definition: Fraction of predicted positives that are truly positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Interpretation:

“When the model says ‘positive’, how often is it correct?”

Important when **false positives are costly** (e.g., spam detection, fraud alerts).



Recall (Sensitivity, TPR)

Definition: Fraction of actual positives that are correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN}$$

Interpretation:

“Of all real positives, how many did the model catch?”

Important when **false negatives are costly** (e.g., disease screening, fault detection).



F1 Score

Definition: Harmonic mean of precision and recall.

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 TP}{2 TP + FP + FN}$$

Interpretation:

Balances precision and recall in a single number.

Useful when classes are **imbalanced** and you care about both FP and FN.



Visual Summary

Metric	Formula	Focus
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness
Precision	$\frac{TP}{TP+FP}$	Quality of positive predictions
Recall	$\frac{TP}{TP+FN}$	Coverage of actual positives
F1 Score	$2 \cdot \frac{P \times R}{P+R}$	Balance between P and R



When to Use Which Metric?

Accuracy

Classes are roughly balanced

All types of errors have similar cost

Precision

False positives are expensive or harmful

You want high confidence in positive predictions



When to Use Which Metric?

Recall

False negatives are expensive or harmful

You want to miss as few positives as possible

F1 Score

Imbalanced classes

You need a single metric reflecting both precision and recall



Evaluating the Performance of a Classifier

Confusion Matrix of LDA predictions on the Default dataset
(10,000 observations)

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000



Problems with Intuition Behind Generative Models

These methods approximate the Bayes Classifier, which aims at minimizing the total classification error (= on all classes!)

For a bank, misclassifying a person who will default is often more critical than misclassifying a person who will not default

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000



How Does the Bayes Classifier Work?

The Bayes classifier assigns an observation to the class for which the posterior probability $p_k(X)$ (or, better, its estimate) is the largest.

For $K = 2$ (binary classification) and $p = 1$, this means to use the threshold 50% on the posterior probability

$$Pr(\text{default} = \text{Yes} | X = x) > 0.5$$



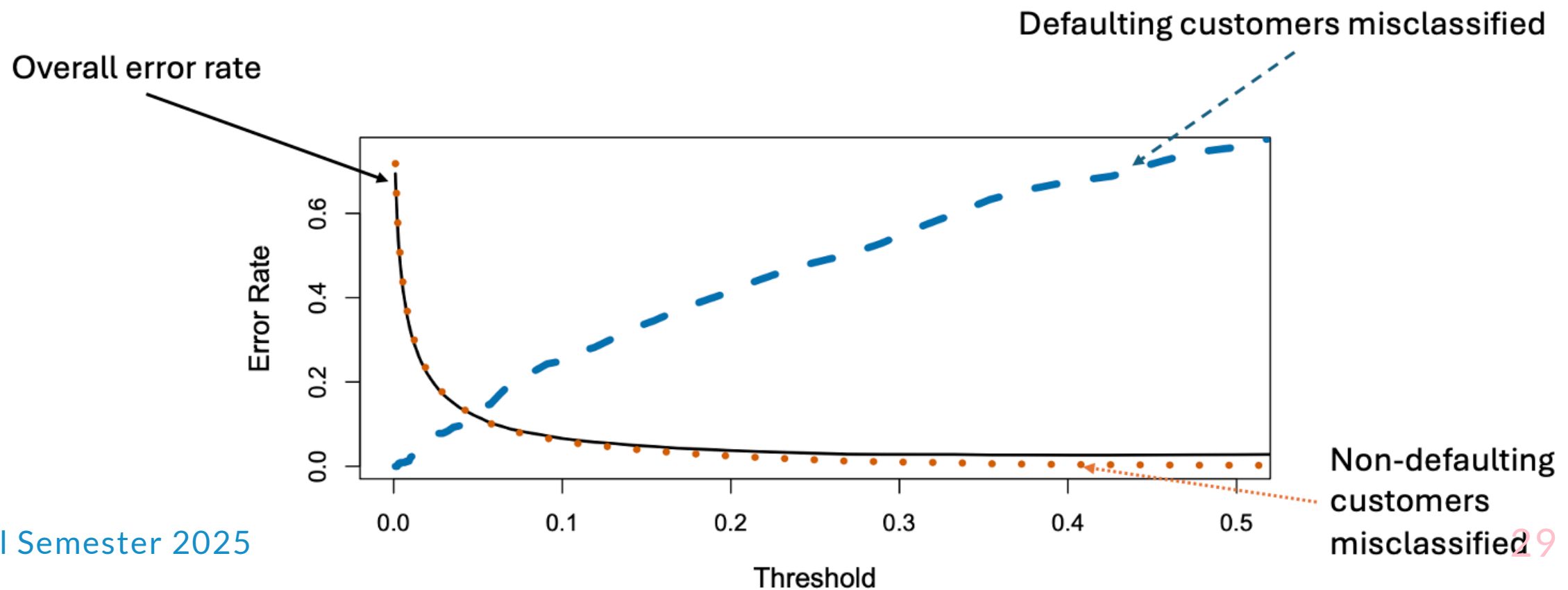
Can we Adjust the Classification Threshold?

Yes, but what does it mean to do it?

$$Pr(\text{default} = \text{Yes} | X = x) > 0.2$$

0.5		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000

0.2		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9432	138	9570
	Yes	235	195	430
	Total	9667	333	10000





Terminology!

		<i>True class</i>		
		– or Null	+ or Non-null	Total
<i>Predicted class</i>	– or Null	True Neg. (TN)	False Neg. (FN)	N*
	+ or Non-null	False Pos. (FP)	True Pos. (TP)	P*
	Total	N	P	

Name	Definition	Synonyms
False Pos. rate	FP/N	Type I error, 1–Specificity
True Pos. rate	TP/P	1–Type II error, power, sensitivity, recall
Pos. Pred. value	TP/P*	Precision, 1–false discovery proportion
Neg. Pred. value	TN/N*	

Default example: + (positive) is the “defaulting” class (customers who cannot repay their CC debt)

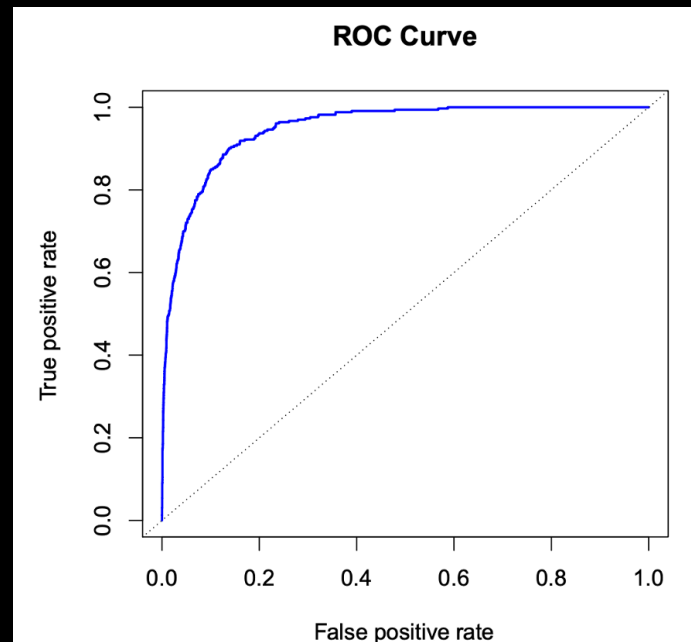


The ROC Curve

ROC: Receiver Operating Characteristics

Plots two types of performance measures as a function of the classification threshold

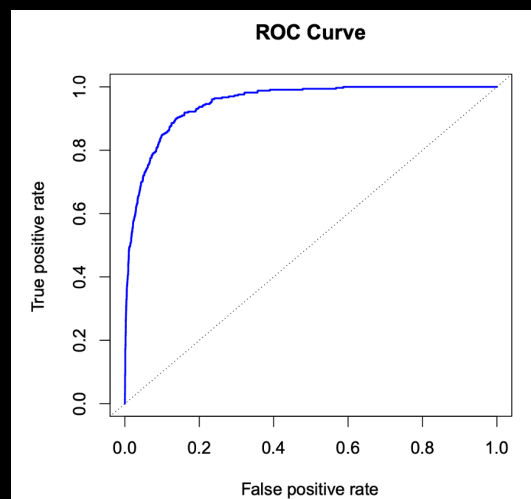
(False positive: non-defaulters classified as defaulters)





When Is a Classifier a Good One?

When its Area Under the ROC Curve (AUC) is close to 1





Poisson Regression

Until now, the task was either *regression* or *classification*

In some cases, it is neither of the two (or somewhere in between)

Example: *Bikeshare* dataset in the textbook,

Response Y counts the hourly number of users of a bikeshare program

Predictors are: month, hour, workingday, temperature, “weather” (clear, cloudy, light rain/snow, heavy rain/snow)

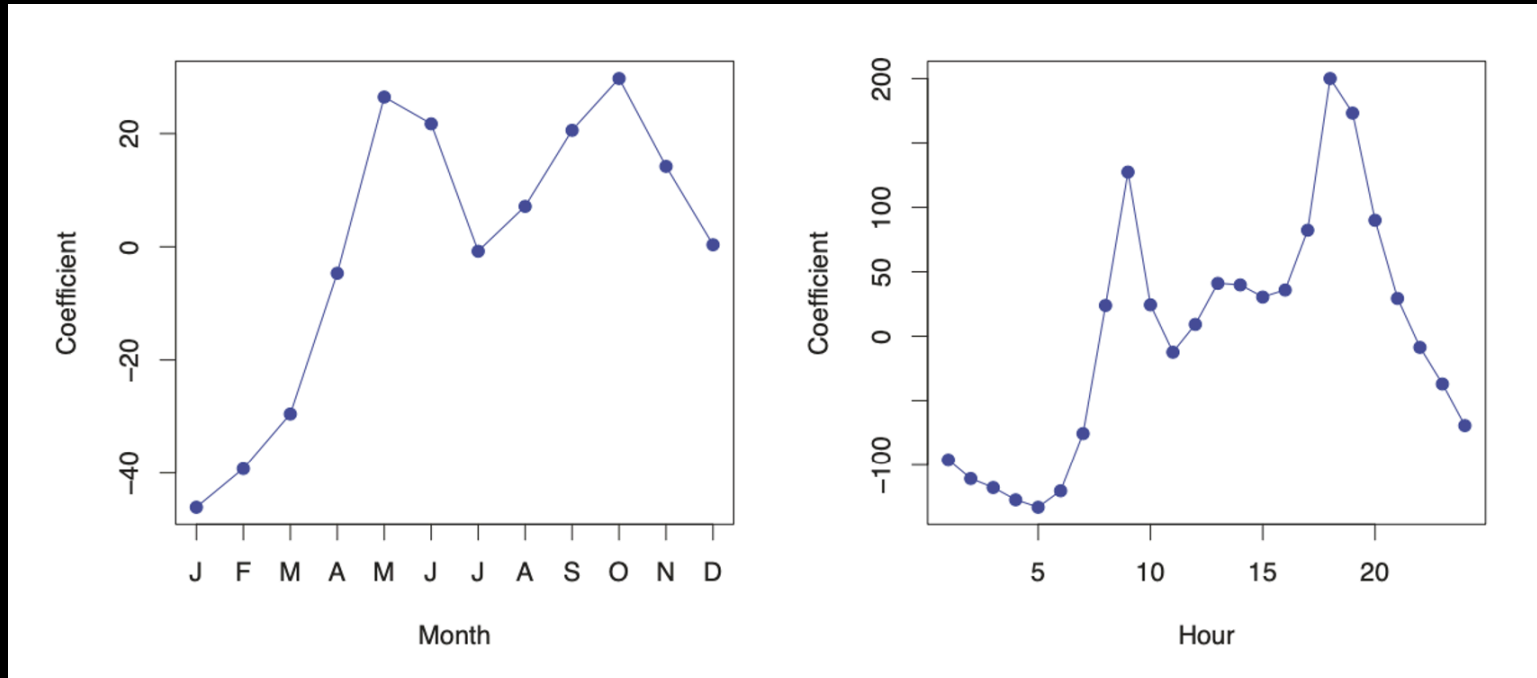


Linear Regression on Bikeshare dataset (p1)

	Coefficient	Std. error	<i>t</i> -statistic	<i>p</i> -value
Intercept	73.60	5.13	14.34	0.00
workingday	1.27	1.78	0.71	0.48
temp	157.21	10.26	15.32	0.00
weathersit[cloudy/misty]	-12.89	1.96	-6.56	0.00
weathersit[light rain/snow]	-66.49	2.97	-22.43	0.00
weathersit[heavy rain/snow]	-109.75	76.67	-1.43	0.15



Linear Regression on Bikeshare dataset (p2)





Linear Regression on Bikeshare dataset

About 10% of the prediction of the linear regression are model are negative

A negative number of bicycle users??

More generally, linear regression does not guarantee that the predicted value will be positive (or comprised between 0 and 1)

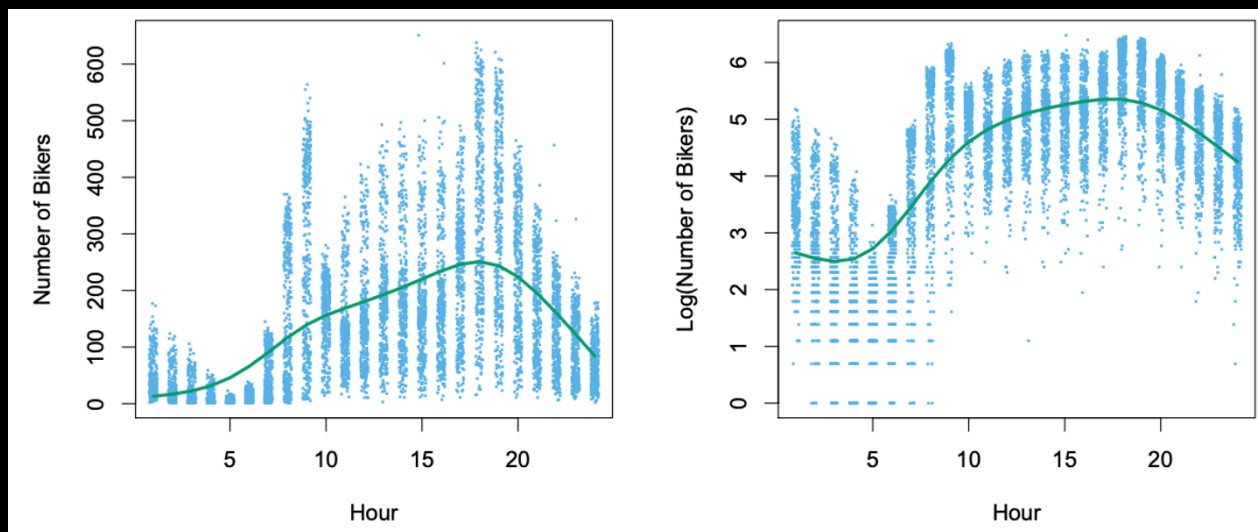


The Problem is Deeper ...

In linear regression $Y = \beta_0 + \sum_{i=1}^p \beta_j X_p + \epsilon$

... with ϵ of mean equal to zero and variance independent of X_p

Is this true in this dataset?





The Poisson Distribution

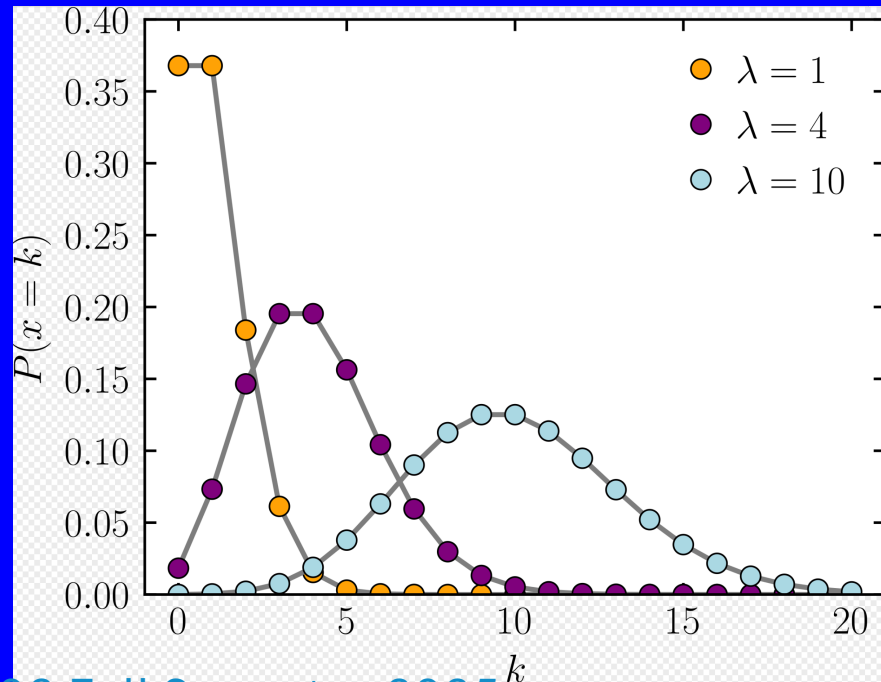
Given a random variable Y with non-negative integer values, Y follows the Poisson distribution if:

$$Pr(Y = k) = \frac{e^{-\lambda} \lambda^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

$$\lambda = E(Y) = Var(Y)$$



Good to model counters (business service systems):
The number of customers arriving at a bank at a certain hour
The number of users of a bike sharing service
(The number of chewing gums on the tiles of a highly beaten pedestrian road)





Poisson Regression

The numbers of users Y of the bikesharing vary as a function of all the predictors $X_1, \dots, X_p$

We want to enforce that $\lambda = E(Y)$ varies as a function of $X_1, \dots, X_p$

That is: $\lambda = E(Y) = \lambda(X_1, \dots, X_p)$

Now assume $\log(\lambda)$ is linear:

$$\log(\lambda(X_1, \dots, X_p)) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

$$\lambda(X_1, \dots, X_p) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$$



Poisson Regression

To sum up:

$$Pr(Y = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$\lambda(X_1, \dots, X_p) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$$

So, $p + 1$ parameters to estimate



Poisson Regression: Parameters Estimation

We use Maximum Likelihood, like for Logistic Regression

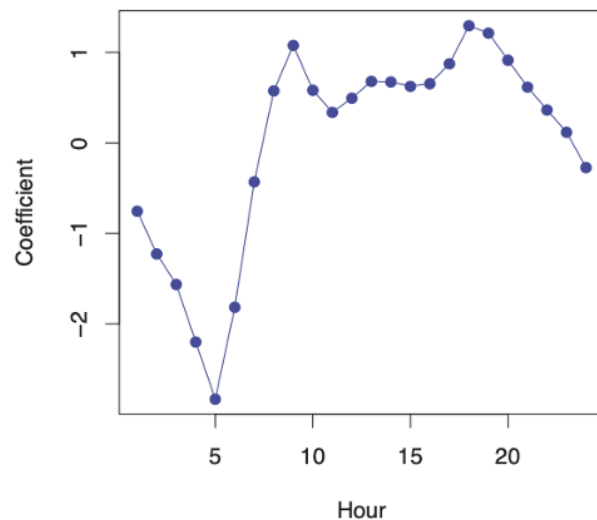
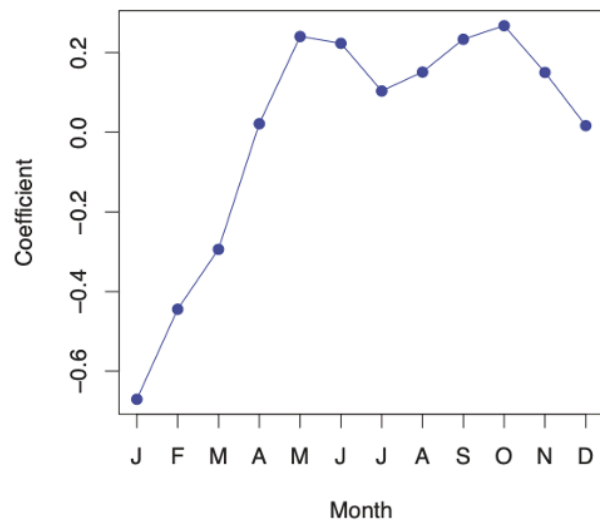
Find the parametr that maximise the likelihood:

$$l(\beta_0, \beta_1, \dots, \beta_p) = \prod_{i=1}^n \frac{e^{-\lambda(x_i)} \lambda(x_i)^{y_i}}{y_i!}$$



Poisson Regression on Bikeshare Dataset

	Coefficient	Std. error	z-statistic	p-value
Intercept	4.12	0.01	683.96	0.00
workingday	0.01	0.00	7.5	0.00
temp	0.79	0.01	68.43	0.00
weathersit[cloudy/misty]	-0.08	0.00	-34.53	0.00
weathersit[light rain/snow]	-0.58	0.00	-141.91	0.00
weathersit[heavy rain/snow]	-0.93	0.17	-5.55	0.00





Interpretation

$$\lambda(X_1, \dots, X_p) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$$

An increase of one unit of X_j is associated with a change of $E(Y) = \lambda$ of a factor e^{β_j}

If it gets cloudy, the mean bike usage will decline of a factor $e^{-0.08} = 0.923$ (that is, on average, 92.3% as many people will use bikes when it is cloudy compared to when it is clear)

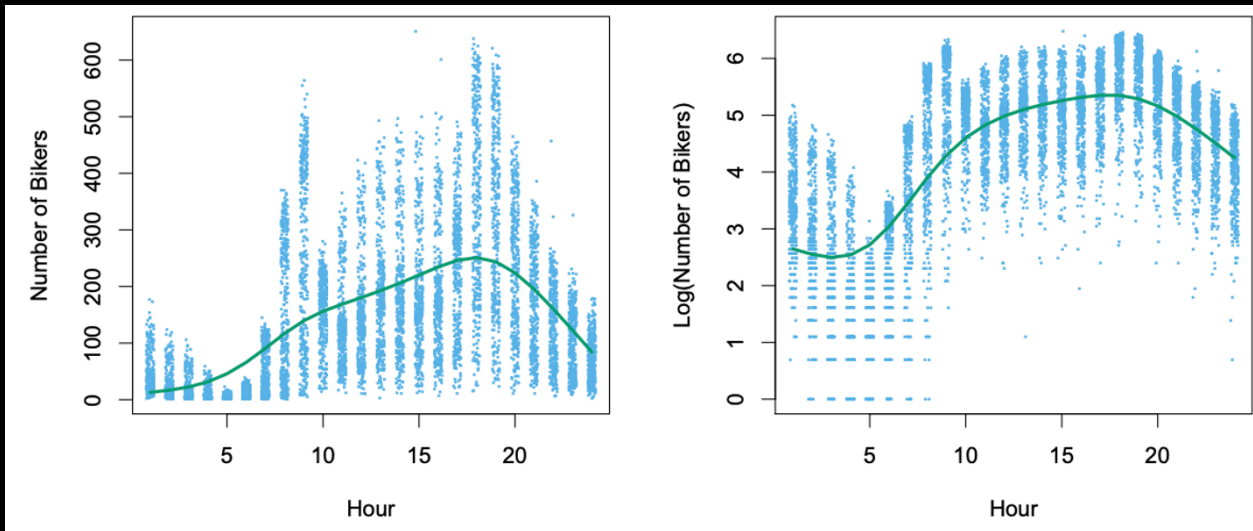
If it rains, the factor becomes $e^{-0.5} = 0.607$ (60% of bike user compared to clear weather)



Interpretation: Mean-Variance Relationship

In the Poisson Regression model: $\lambda = E(Y) = Var(Y)$

Poisson Regression assumes that the variance of the response in a given time frame is equal to the mean in that time frame (not constant!)





Classification: What Did We Learn?

Logistic Regression: a general approach for classification

If the predictors are normally distributed across classes:

LDA (common variance)

QDA

Naive Bayes (independent predictors)

Poisson Regression: when the response is a counter

Prediction Performance of Classifiers: Confusion Matrix and ROC (AUC)